QUANTUMFLOOR

Computational Research Report

Job ID: 3b593670

1. Executive Summary

This report details the execution of the QuantumFloor diagnostic pipeline on dataset 'target'. The system evaluated multiple classical baselines against quantum algorithms to determine the optimal strategy.

2. Dataset Intelligence

Modality:	TABULAR
Total Rows:	1025
Unique Rows:	302
Duplicate Rows:	723
Total Columns:	13
Missing Values:	0.00%
Leakage Risk Detected:	0

Scientific Integrity Warnings

Warning: Duplicate feature groups: 302 | Duplicate rows: 723 | Cross-split duplicate groups: 0 | Target leakage detected: 0 | Identifier leakage detected: 0 | Preprocessing leakage: 0

3. Data Preparation

Train/Test Split:	80/20 Stratified
Preprocessing:	Pipeline (Imputation + Scaling)
Leakage Protection:	Enforced

4. Computational Plan

Original Features:	13
Quantum Embedding Dimension:	8
Qubit Budget:	2
Classical Training Size:	820
Quantum Subset Size:	120

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5. Quantum Execution

Execution Backend: SIMULATED EXECUTION

Ansatz: AngleEmbedding + IsingZZ

Qubits: 2

Circuit Depth: 2

Total Operations: 2

6. Classical vs Quantum Baselines

Model	Type	AUROC	F1	Spec@95	Notes
full_feature_classical	Classical	0.8883	0.8053	0.6211	
logreg	Classical	0.9094	0.8529	0.5789	
svm_rbf	Classical	0.8991	0.8208	0.5684	
random_forest	Classical	0.8667	0.7596	0.4947	
qsvm	Quantum	0.8988	0.8208	0.5368	
pqk	Quantum	0.9089	0.8358	0.6526	
vqc	Quantum	0.9113	0.8365	0.6105	

7. Comparison & Explainability

Best Classical AUROC: 0.9094

Best Quantum AUROC: 0.9113

Delta: 0.0019

Explainability Status: available

PC1: 0.3144

PC2: 0.0706

8. Final Verdict

Best Overall Model: vqc

Quantum Advantage: Yes